Assignment 5: Fundamental Research: Reconfiguring Dominoes on a Square Grid

In this assignment we will investigate gourds, or dominoes, on boards based on a square grid. We will look at what boards always allow reconfiguration to another position of the dominoes. We already know the situation on hexagonal grids with one empty tile: We can often reconfigure. We also know the situation on square grids with one empty tile: we can basically never reconfigure freely. So one next question is: Square grids with two empty tiles.

1 Square grids, boards, and moves

We are interested in boards based on the square grid, composed of a set of squares joined together at their edges to form a board. We consider boards with an even number of squares, and dominoes that fill the board except for two squares, which are empty. So if we have n dominoes, we have $2n+2$ squares. A domino may have one or two colors, that is, the halves may have different colors. This makes the orientation of a domino important. We consider boards that are 2-connected and have no holes. A board is 2-connected if and only if for any two squares on the board, there are two sequences of squares (paths in the dual graph) that do not use the same squares, except for the end squares. A path can have length zero if the two squares we examine are adjacent. Then there needs to be another path.

The moves we allow are the straight slide (in the direction of the domino), and the 90 degree turn (where the empty grid cell is filled by the adjacent part of the domino, and the other part of the domino takes over the position of the part that is moving to the empty grid cell). In certain special situations, other moves can be made, in particular, when a domino and has two adjacent empty squares that together form a 2×2 subgrid. Then a sideways slide can be made, and also a pivot. A pivot is a 90 degree turn when one domino half is stationary. A pivot is not possible if there is only one empty cell in the neighborhood.

We choose the setting with two empty squares, because we already know that boards with an odd number of squares and only one empty square, never allow arbitrary reconfigurations. This is true even for rectangular boards like the 3×3 board with four dominoes.

We will examine the relationship between admitting a Hamiltonian cycle and always being able to reconfigure for such boards. In particular, we will show that if a board (by default 2-connected and hole-free) admits a Hamiltonian cycle, then reconfiguration is always possible. This requires a proof, and proofs have "ingredients". Each ingredient can be stated as a lemma and must be validated with a proof. Some ingredients are simple and a very short proof suffices. Other ingredients are harder and the proof may be longer.

Ingredient 1: Show that for a Hamiltonian cycle on a board as specified, no matter how the n dominoes start, they can always be moved onto the cycle. That is, they each cover two squares that are adjacent in the cycle.

Ingredient 2: Show that every Hamiltonian cycle on a board has a U-configuration: two squares adjacent in the Hamiltonian cycle where the cycle makes a bend in the same direction. The U consists of these two squares plus the two squares adjacent in the Hamiltonian cycle; together they form a 2×2 square.

Ingredient 3: Show that we can always get the two empty spaces adjacent in the U (considering Ingredient 4, where in the U do you want them?).

Ingredient 4: Show that we can always reverse a domino, making use of the U.

Ingredient 5: Show that if we consider six squares of the cycle, namely the four U-squares, and the two before and the two after, and these contain two dominoes and two empty squares, we can always swap the order of the two dominoes in the cycle.

Final step: Show that these ingredients, used in the right way (the "recipe"), together form a proof that any two configurations of dominoes on the board can be transformed into each other.

Further results: Show that there exists a board that is 2-connected and admits a Hamiltonian cycle, but which does not allow arbitrary reconfigurations. Of course this board must have a hole, otherwise our theorem states that reconfigurations are always possible.

Also show that there exists a 2-connected hole-free board with an even number of squares that does not admit a Hamiltonian cycle and does not allow arbitrary reconfiguration.

These two examples show that the two conditions (1) hole-free, and (2) has a Hamiltonian cycle, are conditions we cannot simply leave out. But they are not *necessary* conditions either: there exists a 2-connected hole-free board that allows any reconfiguration but does not have a Hamiltonian cycle. (The smallest example I can find has 20 grid cells.)

2 Assignment

Prepare a plain document that answers the following questions. Do not write a paper, just answer the questions. Use Latex and use illustrations. Illustrations are not a substitute for a proof, but help the reader understand what the text states. A figure can assist a "proof" when it shows all cases that may occur, in which case the text states that all possible cases (ignoring symmetries) are shown in Figure XX. So refer in text to the figures and explain in text what the figure does: is it a listing of all cases, or is it an example that explains intuition? I suggest the use of ipe: `https://ipe.otfried.org/`. You can use Latex to annotate your ipe figures if you link ipe to a Latex compiler. If you save ipe files as pdf, you can directly import them into your Latex source with `\includegraphics{filename}`.

1. State the five lemmas and the five proofs for the ingredients. If you don't have the proof, still formulate the lemma.

2. State a theorem for the real result (the final step), where the ingredients are needed, and write the proof of the theorem, using the lemmas.

You are allowed to look into other papers, but you do not have to. Studying the gourds paper is very useful, but you do not need anything else. You are not allowed to ask for answers in online fora: this is a violation of scientific integrity. You may also not use generative AIs or LLMs in any way.

How do you do such research? Typically, you begin by building up intuition. For Ingredient 1, draw a couple of boards of 16–24 squares, draw a Hamiltonian cycle, take some dominoes or pieces of paper of 2×1, place them on the board, and see how you would do it. Then look at the gourds paper on arXiv (also in the supplied materials); it contains a proof of basically this property on the hexagonal grid with one cell empty. A very similar proof should work here.

For Ingredient 2, the gourds paper also helps. Again we draw several Hamiltonian paths on different boards and observe that a U always appears, and you will fail if you try to avoid it. The idea is to study a graph related to the interior of the Hamiltonian cycle. If we take the grid vertices and grid edges bounding the cells, in particular those that are fully inside the Hamiltonian cycle, then their graph has a useful structure which we can exploit.

For Ingredient 3, see how dominoes move along the Hamiltonian path, and describe how this works and can always give a situation where the empty squares are in the U.

You should now have enough ideas to address Ingredients 4 and 5 as well, and to state and prove the main theorem, the final step.

In most cases, a proof starts by presenting the precise notation (and sometimes defining terminology precisely as well) that will be used in the proof. Only once this is done, the reasoning (proof itself) starts, explicitly using this notation and terminology.

When you work, work as a team. In particular, try to work through the steps in the proof as a team and analyse the cases as a team. This is more fun and it is more likely to avoid errors. When you have the structure of the proof and the arguments, you can distribute work and let different team members write up different proofs. Be sure to check each other's write-up. In science, all authors are responsible for everything in the paper (and not just the part they wrote). If you are unsure of a proof step of someone else, either ask that other person, or try to fill in the missing substep that you need in order to get a more complete proof. When you scrutinize a part of a proof, read a sentence exactly for what it states. If that sentence, literally interpreted, is not quite right, then do something about it. Every sentence of a proof must be precise, fully correct, and unambiguous.

Grading: Grading will depend on (1) precise, correct statements of the lemmas and the theorems, and precise, correct proofs for these results. Also the (2) correct use of notation, the clarity of the arguments, the use of correct English and the layout counts. Finally, the (3) support by clear figures also contributes to the grade; recall that a figure supports the intuition of a definition or a proof, but never replaces it. A figure always needs accompanying text that explains it.

It is not a problem if you fail to make a correct proof, or any proof, for one or two of the lemmas. This assignment is more about precision of notation and argumentation than about quantity of results.

Handing in: By e-mail to the lecturer, Marc van Kreveld (`m.j.vankreveld@uu.nl`). Please cc your whole team. Name the pdf that you submit `MSCIP-5-X.pdf`, where `X` is the number of your team. Make sure that all team members are listed in the pdf, with the tasks they did and number of hours they spent.

Most common mistakes from previous years:

- Groups used the pivot move without assuming the two cells involved are both empty.
- Concepts used in a proof that are not well defined (either wrong or incomplete or missing). One group write "when the dominoes are aligned along a reference line", but never defined what a reference line is.
- Sentences that are unclear. Any sentence must make sense to the reader, not to the writer. In a previous year, a group wrote "If a domino is closed between two others,". I do not understand what that means.
- Sentences that are incorrect when read literally.
- Assumptions made and used that are not fulfilled, leading to an incorrect proof.
- Cases that were forgotten, like when a hand-in has a figure that shows only one of the possible cases (and only this case is treated in text).
- Stating something for true in a proof when this is exactly what needed to be shown.
- Using words that indicate a poor proof, like: "XXX must happen eventually", or "YYY is the case no matter what". The words "eventually" and "no matter what" should not be used in proofs. They ignore the crux of the proof.
- In lemmas and theorems, forgetting to state the assumption that the board is hole-free. You need to include the conditions under which the statement is true, especially for theorems.
- Sloppiness. Last year one group wrote as the caption to a figure: "Main Caption to the Figure".